



Today I am learning to explain, check, and generalise patterns and relationships using algebra.

1. Explain the Rule

Look at the pattern in the table.

Input	1	2	3	4	5
Output	4	6	8	10	12

a) What is the rule?

b) How do you know?

2. Check the Solution

Check each student's answer.

a) A student solved:

$$x + 5 = 12$$

$$x = 7$$

Is the answer correct?

☐ Yes

☐ No

Show how you checked it.

b) A student solved:

$$n - 4 = 9$$

$$n = 13$$

Is the answer correct?

☐ Yes

☐ No

Show how you checked it.

3. Explain the Relationship

Look at the table.

Input	1	2	3	4
Output	3	6	9	12

a) The output is always: _____

b) The relationship is: _____

4. Generalise the Pattern

Look at the growing pattern.



Group 1



Group 2



Group 3



Group 4



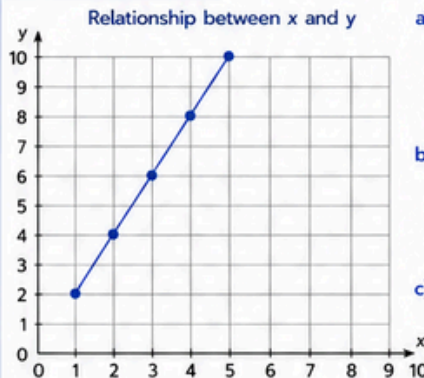
Group 5

a) How many stars would be in Group 10? Answer: _____

b) How did you work it out?

5. Explain the Graph

Look at the graph.



a) What does the graph show?

b) What happens each time x increases?

c) How could you describe the rule?

6. Spot the Error

A student created this table and rule.

Input	1	2	3	4
Output	2	5	8	12

Rule: Add 3

a) Is the rule correct? ☐ Yes ☐ No

b) Explain your answer.



7. Create and Generalise

Create your own pattern.

Group 1



Group 2



Group 3



Group 4



a) What is the rule? _____

b) How could you describe Group 10?

8. Reflect and Explain

Answer in complete sentences.



a) What is the most important thing you learned about patterns and algebra?



b) Why is checking your answer important?



c) How can a rule help predict future values?



I can...

- ☐ explain rules and relationships.
- ☐ check answers using different methods.
- ☐ generalise patterns and describe relationships.
- ☐ use algebra to model and make predictions.



My Learning Goal

I can refine my explanations, check my work, and generalise relationships with confidence.

